Scott Johnson

Contact Information

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Summary

Principal-level Systems Engineer with over 15 years of experience specializing in **C++ engine architecture**, **rendering pipelines**, and **system performance optimization**. Deep expertise in low-level internals (Mozilla Gecko), with a proven history of engineering core graphic systems that underpin modern applications. Passionate about **mentoring** engineering teams, driving code quality standards, and solving complex **multi-threading** and memory management challenges. Currently finishing a PhD in Computer Graphics.

Core Competencies

- **Engine Architecture & Internals**: Deep knowledge of massive C++ codebases (Mozilla Gecko), Layout & Rendering pipelines, and System Architecture.
- **System Performance & Optimization**: CPU/GPU profiling, Memory management, Thread safety strategies, and V8/SpiderMonkey engine optimization.
- **Graphics Programming**: OpenGL, DirectX, Vulkan, and custom shader development (GLSL/HLSL).
- **Leadership & Mentoring**: Leading technical initiatives, mentoring senior engineers, and bridging communication between artists, designers, and engineers.

Work Experience

Remote

Capture Software Engineer at Chromatic **Mar 2023 - Present**

- Architected fault-tolerant distributed systems for visual testing, ensuring high availability and scalability.
- Optimized image processing algorithms using low-level rendering expertise, directly improving performance for developer tools.
- Led cross-team initiatives to improve infrastructure maintainability and build pipelines.
- Performed research into new image comparison algorithms, resulting in implementation of several features unique to Chromatic.

Founder/CEO at FoamFactory **Jan 2021 - Mar 2023**

- Led full-stack architecture of a SaaS platform, balancing technical trade-offs between rapid iteration and system stability.
- Architected a high-performance backend leveraging **Rust**, prioritizing type safety and memory security.

Principal Engineer at Medal **Dec 2019 - Jan 2021**

- **Game Tech Architecture:** Engineered core recording technology for a high-traffic desktop gaming application, working with native node modules and OS-level integrations.
- **Performance Optimization:** Developed a secure, hardware-accelerated video capture system, optimizing low-level resource usage (CPU/GPU) to ensure minimal impact on gameplay frame rates.
- **Cross-Functional Leadership:** Collaborated with engineering and design teams to align on technical strategy and mentored team members on debugging complex threading and performance issues.

Lead Graphics Engineer at InVision App **Dec 2016 - Dec 2019**

- **Rendering Engine:** Designed and implemented a GPU-based vector rendering engine using **WebGL**, **Rust**, **WebAssembly**, **haXe**, and **TypeScript**, solving complex performance challenges to deliver 60fps interactivity.
- **Developer Experience:** Prototyped next-generation rendering solutions using WebAssembly and Rust to increase developer efficiency.

Minneapolis, MN

Mobile Engineering Lead at When I Work **Sep 2014 - Dec 2016**

- Led a team of developers in an Agile environment, instituting rigorous code reviews and CI/CD practices.
- Collaborated with product and design stakeholders to deliver robust application features.

Edina, MN

Android Engineering Lead at Jingit **Nov 2013 - Sep 2014**

- Led several cross-functional teams in developing mobile solutions, focusing on user engagement and application performance.
- Mentored junior engineers and contractors to form a cohesive team focused on building and scaling a retail interaction application on several mobile platforms.

Remote

Platform Engineer, Layout at Mozilla **Jun 2011 - Oct 2013**

- **C++ Engine Internals:** Engineered performance-critical, low-level rendering and layout systems within the **Gecko engine** using **C++**.
- **Standards & Stability:** Architected core rendering systems compliant with W3C standards (CSSOM, WebGL), ensuring cross-platform stability.
- **Performance Profiling:** Optimized rendering pipelines for high-performance web applications, leveraging deep knowledge of **DirectX** and **OpenGL** to maximize hardware utilization.
- Contributed to large-scale open-source projects, collaborating with the global community to drive web technology evolution.

Education

University of Minnesota

Master of Science, Computer Science **2006-2009**

Graduate research focused on computational geometry, photorealistic

rendering, and algorithm optimization. Graduated summa cum laude.

University of North Dakota

Bachelor of Science in Computer Science **2001-2006**

Focused study in software engineering and computer graphics. Graduated summa cum laude.

Bachelor of Science (Mathematics) **2001-2006**

Focused study in statistical theory, number theory, and combinatorics. Graduated summa cum laude.

Awards and Recognitions

- **Open Source Contribution**: Google Summer of Code Mentor & Grant Recipient (Crystal Space 3D SDK), 2007-2009.
- Top Secret Clearance (Currently Inactive); *United States Department of Defense*, 2009-2011.
- Inducted Lifetime Member, *Phi Beta Kappa* & *Upsilon Pi Epsilon*.

Skills

- **Languages**: C++14/17 (High Proficiency), Rust, GLSL/HLSL, TypeScript/JavaScript, Go, Python.
- **Graphics & Systems**: Unreal Engine 5, WebGL, OpenGL, Vulkan, DirectX, Multi-threaded Programming, Engine Architecture.
- **Desktop & Web**: Electron, Native Node Modules, Browser Layout/Rendering Engines.
- **DevOps & Tools**: Git/GitHub, CI/CD, AWS, Docker, Linux/MacOS/Windows Development.